

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
 DURATION: 3 HOURS

SUMMER SEMESTER, 2023-2024
 FULL MARKS: 150

CSE 4801: Compiler Design

Programmable calculators are not allowed. Do not write anything on the question paper.
 Answer all 5 (five) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) What is the major weakness of a recursive-descent parser? Explain how this weakness is resolved in non-recursive predictive parser. 10
(CO2)
(PO1)
 - b) Consider the grammar shown in Equation (1):

$$\begin{aligned} S &\rightarrow Aa \mid bAc \mid dc \mid bda \\ A &\rightarrow d \end{aligned} \quad (1)$$
 - i. Augment the grammar and construct the LR(1) item sets. 10
(CO2)
(PO2)
 - ii. Build the corresponding CLR parsing table. 10
(CO2)
(PO2)
2. a) What is a shift-reduce conflict? Explain how shift-reduce conflict arises for the production $S \rightarrow L = R$ as part of a grammar shown in Equation (2). 10
(CO2)
(PO1)

$$\begin{aligned} S &\rightarrow L = R \mid R \\ L &\rightarrow +R \mid id \\ R &\rightarrow L \end{aligned} \quad (2)$$
 - b) Describe various error recovery strategies during syntax analysis. 10
(CO4)
(PO1)
 - c) Write a Yacc code snippet that can convert a given postfix expression into an infix expression. The set of operators supported by the converter is $\{ +, -, /, ^ \}$ with their standard precedence and associativity. Assume that the associated lex program already exists which can recognize numbers, identifiers and operators. 10
(CO2)
(PO1)
3. a) Design a Syntax-Directed Definition (SDD) for the grammar shown in Equation (3) to generate syntax-tree. Assume that `makeleaf()` and `makenode()` are two functions to generate leaf-node and internal-note of the tree and return the node address. 10
(CO2)
(PO3)

$$\begin{aligned} E &\rightarrow E + T \mid T \\ T &\rightarrow T * F \mid F \\ F &\rightarrow (E) \\ F &\rightarrow id \end{aligned} \quad (3)$$
 - b) What is L-Attributed definition? Explain the importance of L-Attributed definitions for efficient evaluation of attributes. 10
(CO2)
(PO1)

- c) Explain how inherited attributes can be evaluated in bottom-up parsing with a suitable mini example. 10
(CO2)
(PO1)
4. a) Differentiate static, stack, and heap allocation for functions. 10
(CO4)
(PO1)
- b) Discuss symbol table implementation issues during compiler design. 10
(CO1)
(PO1)
- c) Syntax-Directed Definition (SDD) for a sample set of expressions is shown in Table 1. Write down the Three-Address Code (TAC) generated for expression $a = b + (c + d)$ using this SDD. 10
(CO3)
(PO2)

Table 1: SDD for Question 4.c containing semantic rules to generate TAC for expressions

Production	Semantic Rules
$S \rightarrow id := E$	$S.code = E.code \parallel gencode(top.get(id.lexeme) \neq E.addr)$
$E \rightarrow E1 + E2$	$E.addr = new Temp()$ $E.code = E1.code \parallel E2.code \parallel$ $gencode(E.addr \neq E1.addr \neq E2.addr)$
$E \rightarrow -E1$	$E.addr = new Temp()$ $E.code = E1.code \parallel$ $gencode(E.addr \neq 'uminus' E1.addr)$
$E \rightarrow (E1)$	$E.addr = E1.addr$ $E.code = E1.code$
$E \rightarrow id$	$E.addr = top.get(id.lexeme)$ $E.code = ''$

5. a) Compare static checking vs. dynamic checking. 4
(CO2)
(PO1)
- b) List different types of static checking along with definitions. 4
(CO2)
(PO1)
- c) Discuss type checking of statements, functions, and arrays. Also provide semantic rules to evaluate type attribute for each of them. 12
(CO2)
(PO1)
- d) List the common forms of three-address statements with brief description. 10
(CO3)
(PO1)

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SUMMER SEMESTER, 2023-2024
 FULL MARKS: 150

CSE 4835: Pattern Recognition

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Justify your agreement/disagreement with the following statements:

3 × 5
 (CO2)
 (PO2)

- i. During the training process, if the loss curve is observed to plateau at a relatively high value in the early stages, one possible refinement is to increase the learning rate.
- ii. One theorem says, "A Neural Network with one hidden layer can approximate any function $f : R^N \rightarrow R^M$ with arbitrary precision." The same theorem is applicable for a Convolutional Neural Network (CNN) with one hidden layer.
- iii. Assume that, in a classification problem, there are two classes: Square and Circle. Figure 1 represents the data points of the two classes. Here, each axis represents one feature. The position of the data point along that axis represent the magnitude of the feature for that particular datapoint. The training set is represented by filled (grey) markers, while the test set is represented by unfilled markers. In this scenario, a linear SVM can achieve 100% accuracy on both the training and test sets.

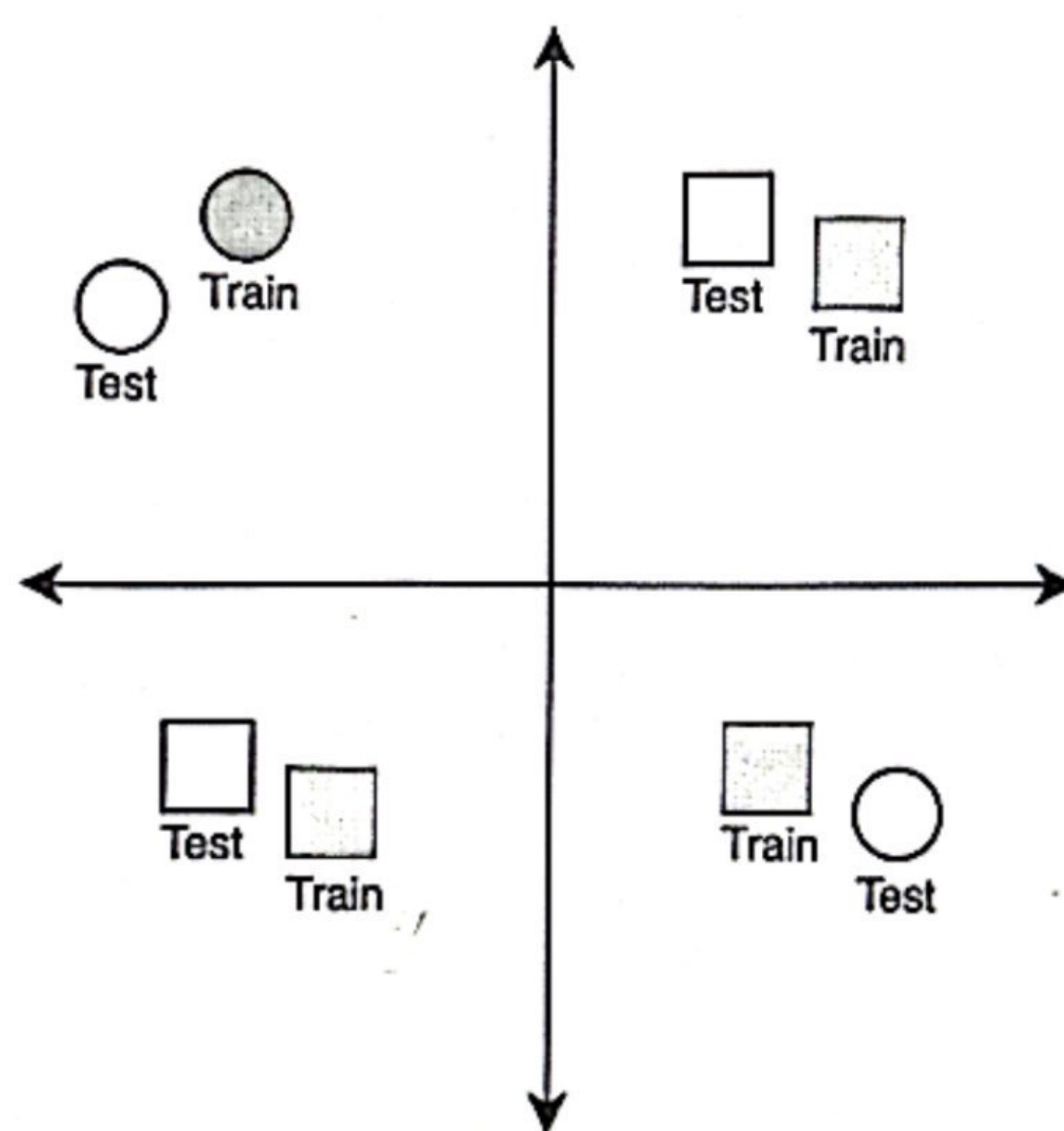


Figure 1: Sample datapoints of two classes for Question 1.a (iii)

- iv. After applying max pooling with spatial extent 2, stride 2, and zero padding on an input with $W \times H \times C$, (where W , C , and H are all divisible by 2), the number of values in the output shape is one eighth that of the input.
 - v. ReLU activation is better than Sigmoid because it is zero-centred.
- b) Consider the following keywords: {Score, Weight vector, Gradient Descent, Loss Function, Input data, Backpropagation, Regularization}
- Draw a flowchart by arranging the aforementioned keywords according to their roles in solving a classification problem. Discuss their roles and relations with each other in brief.

5 + 5
 (CO1)
 (PO1)

2. a) Figure 2.A shows a loss curve produced by a 'black box optimizer'. Figure 2.B and 2.C show the state of the loss curve after tuning a hyperparameter.
- Which hyperparameter is likely to be modified here?
 - Out of the Experiments A, B, and C, which one corresponds to the largest magnitude of the hyperparameter? Justify.
 - The loss curve for Experiment C seems to be the most desirable. Despite this, is there any reason for which someone would choose the hyperparameter in Experiment B for training a model?

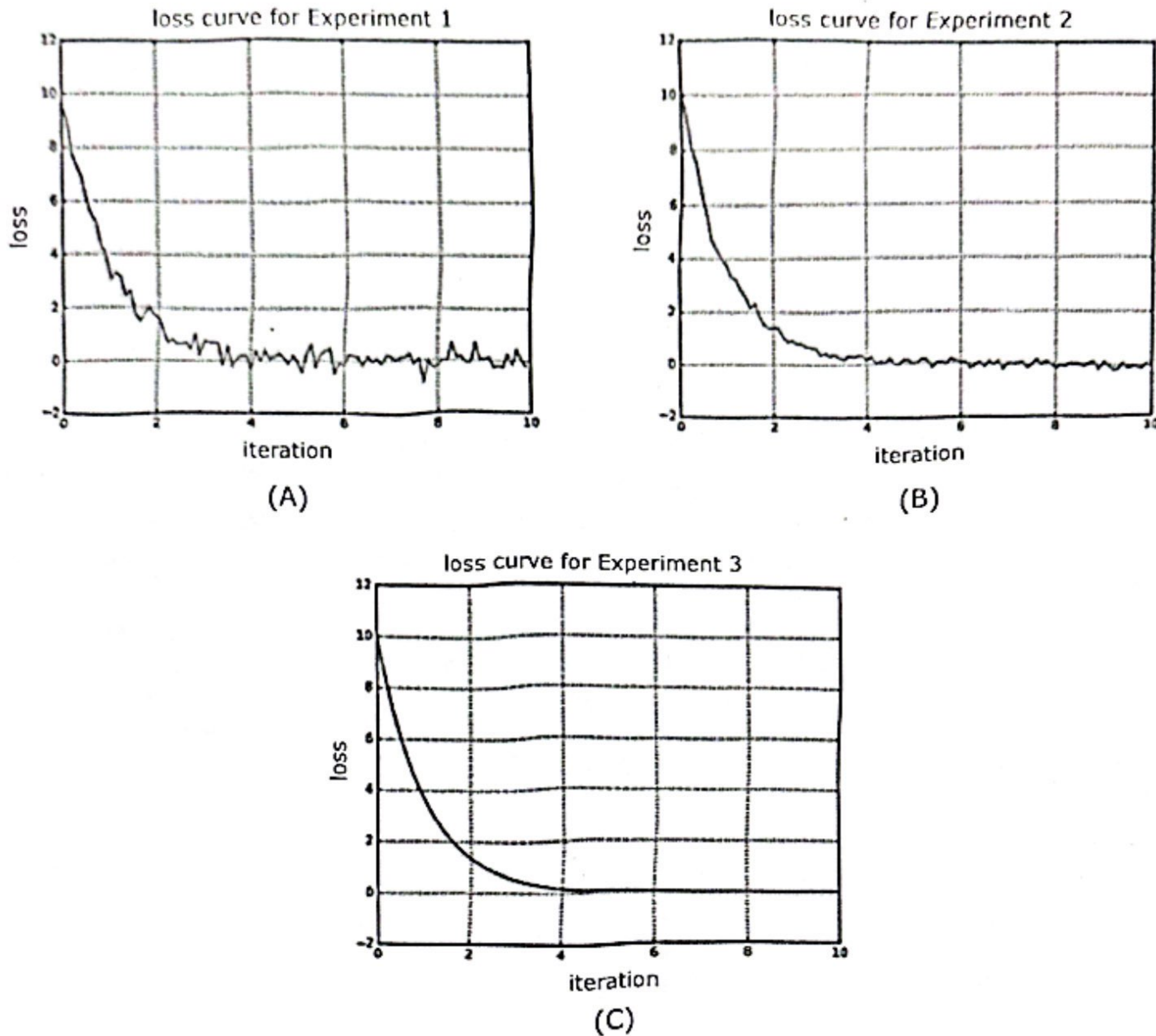


Figure 2: Loss curves of Experiments A, B, and C for Question 2.a

- b) You have a linear classifier with weights W as follows:

$$W = \begin{bmatrix} 3 & -1 & -2 \\ -3 & 1 & 4 \end{bmatrix}$$

This classifier has been trained to classify data points of the form $x = [a, b, c]$, where a represents how spiky a shape is, b represents how large a shape is, and c represents how many dots the shape has.

The classifier will classify the shape as either a 'Wob' (class 1) or a 'Bob' (class 2). So the classifier works as follows:

$$Wx = \begin{bmatrix} \text{'Wob' Score} \\ \text{'Bob' Score} \end{bmatrix}$$

4 + 4
(CO2)
(PO2)

- i. Classify the data point $x = [1, 3, -1]$.
- ii. Assume that the given weights (W) can correctly capture the features of the true objects. Based on the weights, determine class labels of the two shapes given in Figure 3 with appropriate justification.

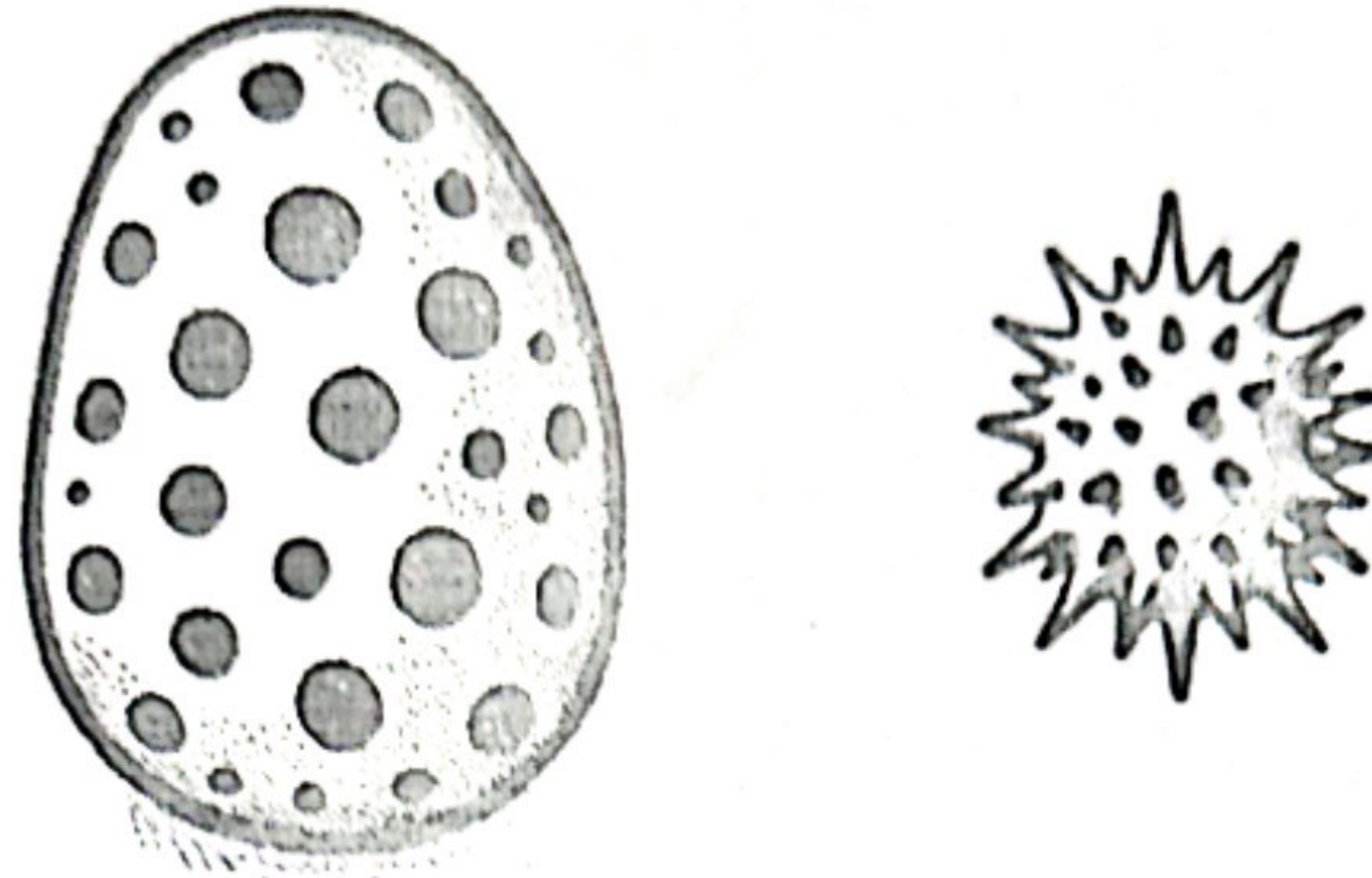


Figure 3: Data Sample for Question 2.b

- c) Model ensembling usually involves training multiple independent models and then providing a final decision based on probability averaging or a max-voting strategy. Suppose you cannot afford the computational resource to train multiple models. How can a cyclic learning rate scheduler be utilized in this situation to avoid multiple model training? 5
(CO1)
(PO1)
3. a) Explain the working procedure of the Fast R-CNN model in detail. 10
(CO1)
(PO1)
- b) A retail company wants to develop a pattern recognition system to automatically classify products based on images taken in their warehouses. Initially, the team trains a Convolutional Neural Network (CNN) from scratch using 20,000 labeled product images. However, the model shows poor generalization when tested on new product categories and images taken under different lighting conditions. Collecting and annotating a larger dataset is costly and time-consuming. 5 × 3
(CO2)
(PO2)
One of the senior engineers suggests applying transfer learning. The idea is to take a CNN pretrained on a large-scale dataset such as ImageNet and then fine-tune it on the company's product dataset. Another engineer argues that feature extraction with frozen layers may be more efficient, given the limited compute resources. Meanwhile, the project manager is concerned about the trade-off between model accuracy and training time, since the solution must be ready within a strict deadline.
Based on the given scenario, answer the following questions:
 - i. Explain how transfer learning could help in this scenario compared to training a model from scratch.
 - ii. Differentiate between fine-tuning and feature extraction with frozen layers in the context of transfer learning. Which approach would you recommend here, and why?
 - iii. Discuss the potential limitations or risks of applying transfer learning in this case.

4. a) Consider Figure 4 to answer the following questions:
- Discuss how the design choices considered inside the Squeeze-and-Excitation block (SE-block) enhances the performance of a vanilla Bottleneck Residual Block.
 - Analyze the computational impact in terms of FLOPs count due to the inclusion of the SE-block.
 - Criticize the idea of replacing the Sigmoid activation in the SE-block with ReLU.

5 ×
(CO2,
(PO2)

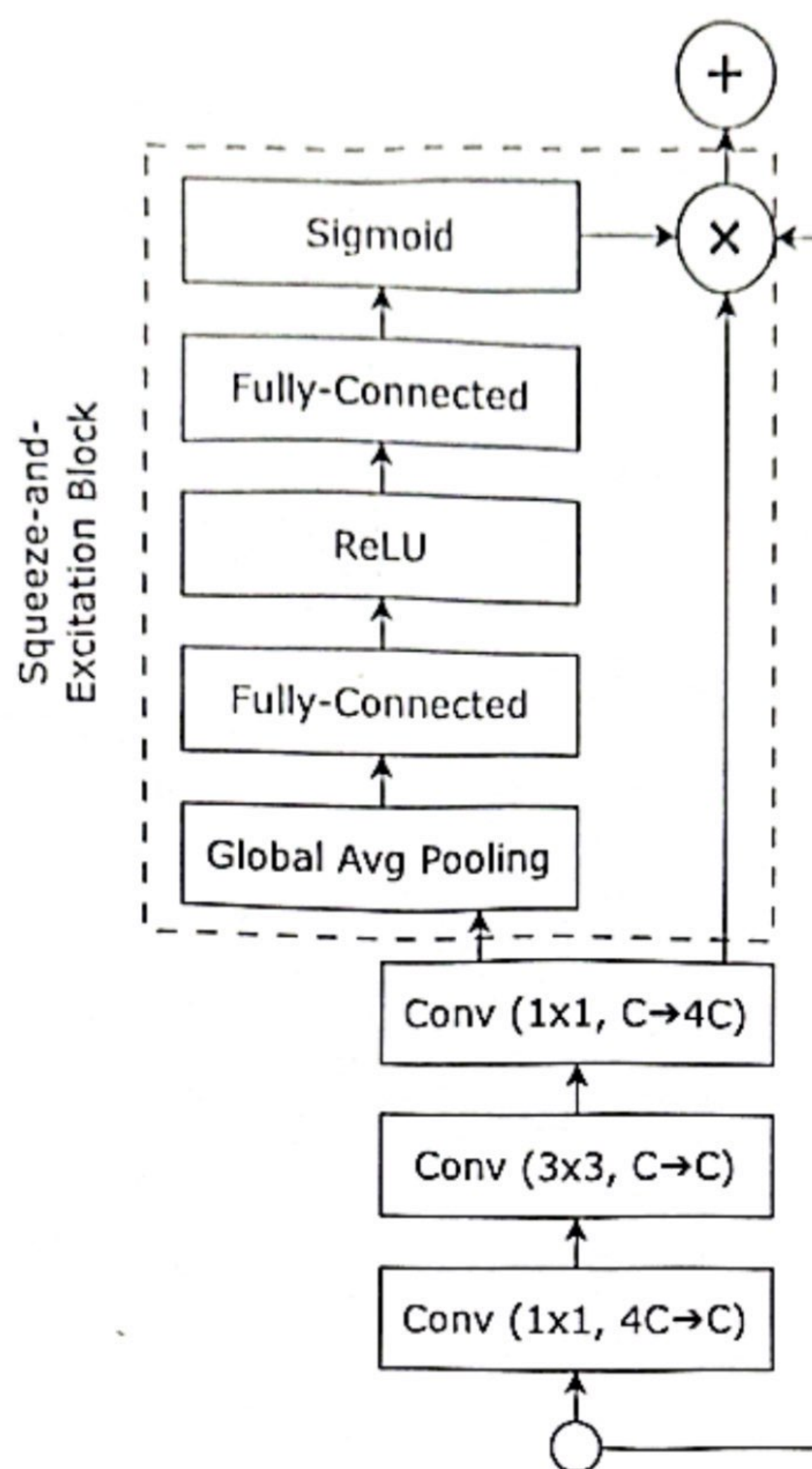


Figure 4: Bottleneck Residual Block with Squeeze and Excitation Module for Question 4.a

- With necessary diagrams, discuss the limitations of building Convolutional Neural Networks (CNNs) via Stacking Grouped Convolution and how ShuffleNet addresses this issue.

5 + 5
(CO1)
(PO1)
5. a) 'MobileNets run extremely fast on CPUs, but it may run slower on GPUs' - Justify the statement.

5
(CO1)
(PO1)
- Discuss the role of the Non-Max Suppression algorithm in object detection, along with its limitations.

3 + 2
(CO1)
(PO1)
 - Explain the detailed working mechanism of a Feature Pyramid Network (FPN) in the context of object detection. Furthermore, discuss how the underlying principles of FPN can be adapted for enhancing image classification backbones and under what circumstances such an adaptation may be beneficial.

10 + 5
(CO1)
(PO1)

6. A group of young researchers is developing a robust pipeline for plant disease classification from leaf images. The dataset contains 18,000 samples belonging to 10 classes. However, class imbalance exists, as one class contains approximately 5,000 samples, whereas another class has only 200 samples. To address this issue, they considered three strategies: upsampling, downsampling, and a hybrid approach. In the upsampling strategy, all other classes are augmented to match the highest sample count. Conversely, the downsampling strategy reduces the number of samples in the majority classes to achieve balance. The hybrid approach applies moderate downsampling to the majority classes while simultaneously employing augmentation-based upsampling for the minority class.

After selecting one of these balancing strategies, the researchers employ a suitable backbone network and train the model for 1,000 epochs with a learning rate of 0.001. Additionally, to further enhance model performance, several dense layers are incorporated after the flattened activation map from the backbone.

Based on the given scenario, answer the following questions:

- | | |
|---|---------------------|
| a) Criticize the three strategies utilized to tackle the dataset imbalance issue. | 5
(CO2)
(PO2) |
| b) Instead of adopting one of those three strategies, propose an alternative approach. | 5
(CO3)
(PO3) |
| c) Criticize the decision of backbone training with 1000 epochs with a learning rate of 0.001, and propose a better strategy. | 5
(CO2)
(PO2) |
| d) Highlight the potential usefulness of the dense layers incorporated at the end of the backbone and some design guidelines to mitigate the risk of overfitting. | 5
(CO3)
(PO3) |
| e) From the different techniques learned in this course, recommend additional strategies that could potentially improve model performance in this context. | 5
(CO3)
(PO3) |

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SEMESTER FINAL EXAMINATION
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SUMMER SEMESTER, 2023-2024
 FULL MARKS: 150

CSE 4849: Human Computer Interaction

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Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
1. a) Suppose you are designing a mobile banking app and considering a specific task, "Transferring money to another account," to analyze five design principles of interaction design (visibility, feedback, consistency, constraints, and affordance).
 - i. For each of the five design principles, write one sentence explaining how it applies to the money transfer task. 10
(CO3)
(PO3)
 - ii. Match each principle with the most relevant usability goal from the list: Effectiveness, Efficiency, Safety, Learnability, Memorability. Write one sentence justification for each goal. 5
(CO3)
(PO3)
 - iii. Explain in two sentences which user experience goals (one or more) from the list: satisfaction, enjoyment, trust, engagement, will be improved when these design principles are applied to the money transfer task. 5
(CO3)
(PO3)
 - b) In the context of conceptualizing interaction, an interactive system can be analyzed based on assumptions and claims. Write two assumptions about the user or system for the money transfer task, and two claims about what the system or design principles will ensure for the user while performing that task. 5
(CO3)
(PO1)
 2. a) What do you understand by divergent thinking and convergent thinking in the design process of a system using the double diamond model? Explain with a real-life example. 15
(CO3)
(PO1)
 - b) Social media app allows users to post updates, interact with friends, explore content, and receive notifications. For each task below, identify the interaction type (instructing, conversing, manipulating, exploring, responding) and justify your choice in one sentence. 10
(CO3)
(PO1)
 - The user taps the Like button on a friends post.
 - The app sends a notification when someone comments on the users post.
 - The user scrolls through their feed to see new posts.
 - The user searches for a friend by typing their name in the search bar
 - The user sends a message to a friend using the chat feature.
 3. a) Imagine you are designing a GPT-based educational assistant that helps university students prepare for exams. The system should collaborate with students by providing answers, explanations, and study guidance while ensuring students remain in control of their learning. Explain how the phases of user-centered design (UCD) can be applied, considering the inclusion of stakeholders in this system. 15
(CO3)
(PO3)
 - b) You are analyzing the design of an e-book reader system. The app supports several user tasks that are designed using different interface metaphors. For each of the following actions, identify the interface metaphor applied, and justify your choice in one sentence. 10
(CO3)
(PO3)
 - A reader quickly jumps to the occurrence of a particular word inside the text without reading the whole book.

- The system displays a collection of available books in a grid, and the reader taps one to start reading.
 - A user marks a passage of interest so they can return to it later or share it with others.
 - When a reader logs into the app from a new device, all previously purchased and read books appear automatically.
 - The reader expands a complex diagram on the page to view fine details more clearly.
4. a) To design a GPT-powered educational assistant, write the persona construction process based on the following steps. 15
(CO4)
(PO1)
- Identify behavioral variables
 - Map interview subjects to behavioral variables
 - Identify significant behavior patterns
 - Synthesize characteristics and relevant goals
 - Check for redundancy and completeness
 - Expand description of attributes and behaviors
 - Designate persona types
- b) Explain the relationship among persona, scenario, and goal with a real-life example. 10
(CO3)
(PO1)
5. a) You are designing a Bangla mental health conversational AI system to provide empathetic support for the stakeholders. Answer the following:
- i. Identify three suitable data gathering methods, explain why each is appropriate, and indicate whether it collects qualitative or quantitative data. 9
(CO4)
(PO1)
 - ii. For each method, describe how you would record and analyze the data to inform system design, giving one example per method. 6
(CO4)
(PO1)
 - iii. Discuss two ethical considerations when collecting sensitive mental health data and how you would address them. 5
(CO4)
(PO1)
- b) Differentiate between 'Evolutionary' prototyping and 'Throw-away' prototyping construction process. 5
(CO4)
(PO1)
6. a) You are evaluating a Bangla mental health support chatbot. Participants must complete two fixed tasks with the chatbot:
- Task 1: "Tell the chatbot you are feeling stressed."
 - Task 2: "Explain to the chatbot that you are feeling a bit lonely and ask it to suggest self-care activities."
- Two chatbot versions are tested: Version A (Empathetic) uses supportive and validating language; Version B (Neutral) provides only factual responses.
Measures collected: completion time, error count, user satisfaction (1-5 scale), trust rating (1-5 scale).
- i. Identify the independent, dependent, control, confounding, and random variables in this empirical study. Give one line justification for each variable you have chosen. 10
(CO5)
(PO3)
 - ii. To compare two chatbot versions: Empathetic (supportive) vs Neutral (factual), decide whether a within-subjects experiment or a between-subjects experiment is more appropriate for this study. Justify your answer in terms of causal effect analysis and correlation analysis. 10
(CO5)
(PO3)
- b) For the inspection-based evaluation of the mental health chatbot, which evaluation technique, between cognitive walkthrough and pluralistic walkthrough, would be appropriate? Justify your choice. 5
(CO5)
(PO3)

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SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 100

CSE 4809: Algorithm Engineering

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 5 (five) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
- | | | |
|----|--|---------------------|
| 1. | a) Prove that the expected number of comparisons in a quick-sort is $n \lg(n)$ | 4
(CO1)
(PO1) |
| | b) Write an algorithm to find the median in expected linear time. Prove its asymptotic complexity. | 8
(CO1)
(PO1) |
| | c) Write a 2-approximation algorithm for traveling salesman problem. Describe the assumptions and prove the optimization ratio. | 8
(CO3)
(PO2) |
| 2. | a) Why is it assumed that shortest path finding problem in P where as longest path finding problem still in NP ? | 5
(CO3)
(PO2) |
| | b) What is the simplified assumption in shortest path finding problem that enable Dijkstra's greedy algorithm? State the correctness of the algorithm. | 5
(CO1)
(PO1) |
| | c) There is a recent (year 2025) improvement of Dijkstra's algorithm in parse graph. What is the improvement? How does the algorithm improve the complexity? | 5
(CO2)
(PO2) |
| | d) Edge relaxation is the mechanism employed in shortest path algorithms in different ways. How edge relaxations are done on Bellman-ford, DAG-based techniques? | 5
(CO1)
(PO1) |
| 3. | a) How does GPU enable Neural Networks in exploiting parallelism? | 5
(CO4)
(PO3) |
| | b) RNN is inherent sequential in design, explain. | 5
(CO4)
(PO3) |
| | c) How does transformer architecture overcome the sequential calculation barriers of RNN? | 5
(CO4)
(PO4) |
| | d) Draw an example computation call-graph and demonstrate a forward pass and backward pass over the graph. | 5
(CO4)
(PO4) |
| 4. | a) What is PoW in Blockchain? How is it Byzantine Fault Tolerant? | 5
(CO2)
(PO2) |

- b) How does a miner win a block under PoW scheme? 5
(CO2)
(PO2)
- c) What is the parallelism in PoW challenge exploited by the mining consortium? 5
(CO3)
(PO2)
- d) How does Merkel-Tree, Merkel Hash, Block Hash and others ensure the integrity of a block in blockchain? 5
(CO2)
(PO2)
5. a) Derive the following equation in Reinforcement Learning (RL), where notations signify their usual meanings. 5
(CO1)
(PO1)
- $$Q_{k+1} = Q_k + \frac{1}{k}[R_k - Q_k]$$
- b) "Exploration and Exploitation" is the mechanism applied to gain overall higher total reward in RL. How do greedy and ϵ -greedy strategy exploit the mechanism. 5
(CO2)
(PO2)
- c) How does the B+ - Tree maintain the balance in tree depth. 5
(CO2)
(PO2)
- d) Write down the essential differences between memory-based balance tree and B-Tree. Why are those differences necessary? 5
(CO1)
(PO1)

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SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 150

CSE 4803: Graph Theory

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Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Given an arbitrary digraph, explain the criteria based on which you can confirm that it contains a directed Euler line. 6
(CO1)
(PO1)
- b) Suppose a biotech company is designing a DNA sequencing machine whose probes can detect a sequence of exactly 3 nucleotides together, and in this simplified model, we assume nucleotides can only be of three types: Adenine (A), Guanine (G), and Cytosine (C). To test the machine, the engineers want to design a circular strand of synthetic DNA such that every possible 3-nucleotide sequence of A, G, and C appears exactly once as a contiguous subsequence when the strand is read. In order to complete this task, the engineers are interested to find out the maximum possible length of the circular strand.
 - i. Formulate the given scenario as a digraph and explain the meaning behind each of the vertices, the edges, as well as the labeling convention you have followed. 11
(CO2)
(PO2)
 - ii. Using the formulation in Question 1.b)i, show how you can construct one such circular strand. 8
(CO3)
(PO3)
2. a) Refer to the graph shown in Figure 1 and answer the following questions.

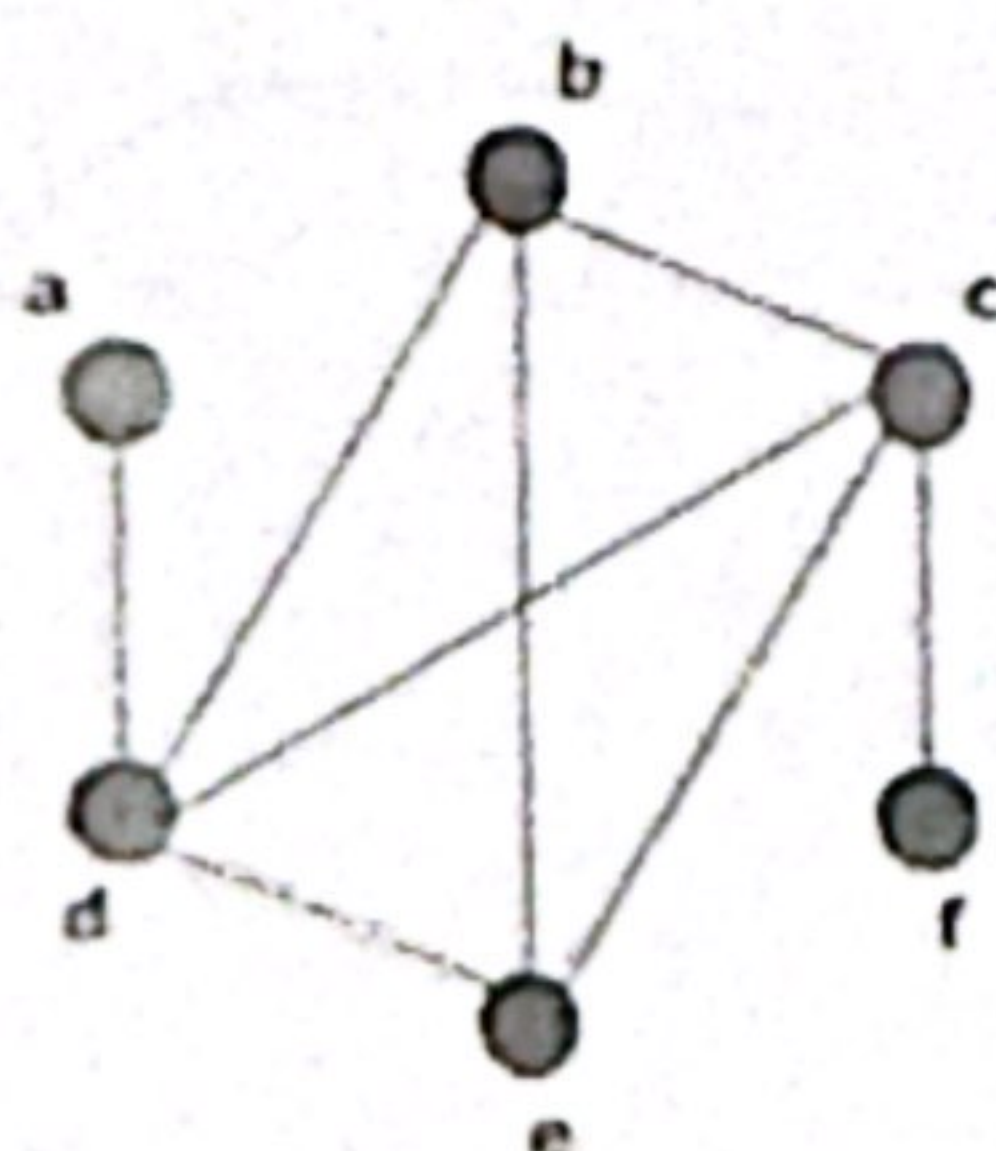


Figure 1: Sample graph for Question 2.a.

- i. Using Boolean Arithmetic method, find all the maximal independent sets of the graph. 12
(CO1)
(PO1)
- ii. Determine the Independence Number of the given graph. 3
(CO1)
(PO1)
- iii. Find all possible chromatic partitionings of the graph. 4
(CO1)
(PO1)
- b) Show a tree in which its diameter is not equal to twice its radius. Explain the conditions under which this inequality holds. 5+3
(CO1)
(PO1)

3. a) Deduce the properties of the labelled trees having the following type of Prüfer codes: 3 × 3
(CO1)
(PO1)
- Prüfer code with a single value.
 - Prüfer code with exactly two values.
 - Prüfer code with distinct values in all positions.
- b) You are given the task of designing the backend logic for a calculator device that will take a string consisting of variables, constants and operators as input and produce the evaluated result of the expression as output. The input will be given as an infix expression with or without parenthesis. The calculator should be able to evaluate the expression efficiently and unambiguously. To test your design, you are given the following expression:
- $$3 + 5 \times 2 - 2^2/2$$
- Assuming the standard PEMDAS rule of operator precedence, convert the given expression into an arborescence. As an example, applying PEMDAS rule on the expression $3^2/2$ is supposed to produce the result 4.5, and not 3, since exponentiation must be done before division. 6
(CO2)
(PO2)
 - Explain why your obtained arborescence is also an *Ordered Tree*. 4
(CO1)
(PO1)
 - Show the evaluation process of the given expression using the obtained arborescence. 6
(CO1)
(PO1)

4. Figure 2 shows a graph G where the bold solid edges labeled b_1 through b_6 constitute one of the many spanning trees of G . The thinner dashed edges labeled c_1 through c_8 represent the chords with respect to this particular spanning tree.

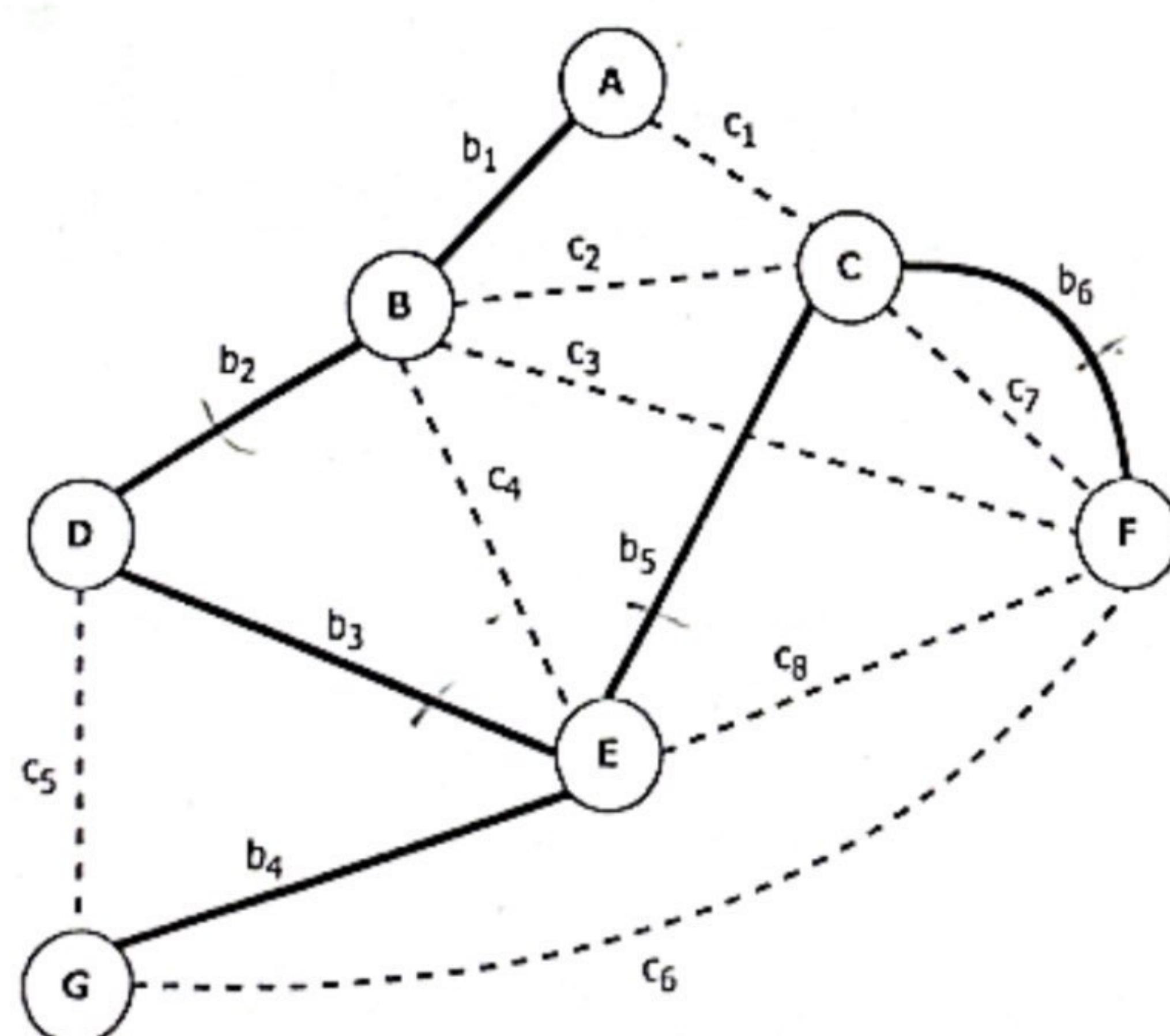


Figure 2: A sample graph for Question 4 where the bold edges represent one of the possible spanning trees.

- Find a new spanning tree of this graph at a distance of 4 from the given spanning tree. Show each of the intermediary spanning trees you generated at distances 1 to 4. 11
(CO1)
(PO1)
- Find all the fundamental circuits with respect to the new spanning tree obtained in Question 4.a. 8
(CO1)
(PO1)
- Find all the fundamental cut-sets with respect to the new spanning tree obtained in Question 4.a. 6
(CO1)
(PO1)

5. Figure 3 shows a bipartite graph that models the eligibility relationship of four job applicants, represented by the 4 vertices in the vertex set V_1 , namely w, x, y , and z , and four available positions, represented by the 4 vertices in the vertex set V_2 , namely a, b, c , and d . An edge from a vertex in V_1 to a vertex in V_2 signifies that the corresponding applicant is eligible for that particular job post.

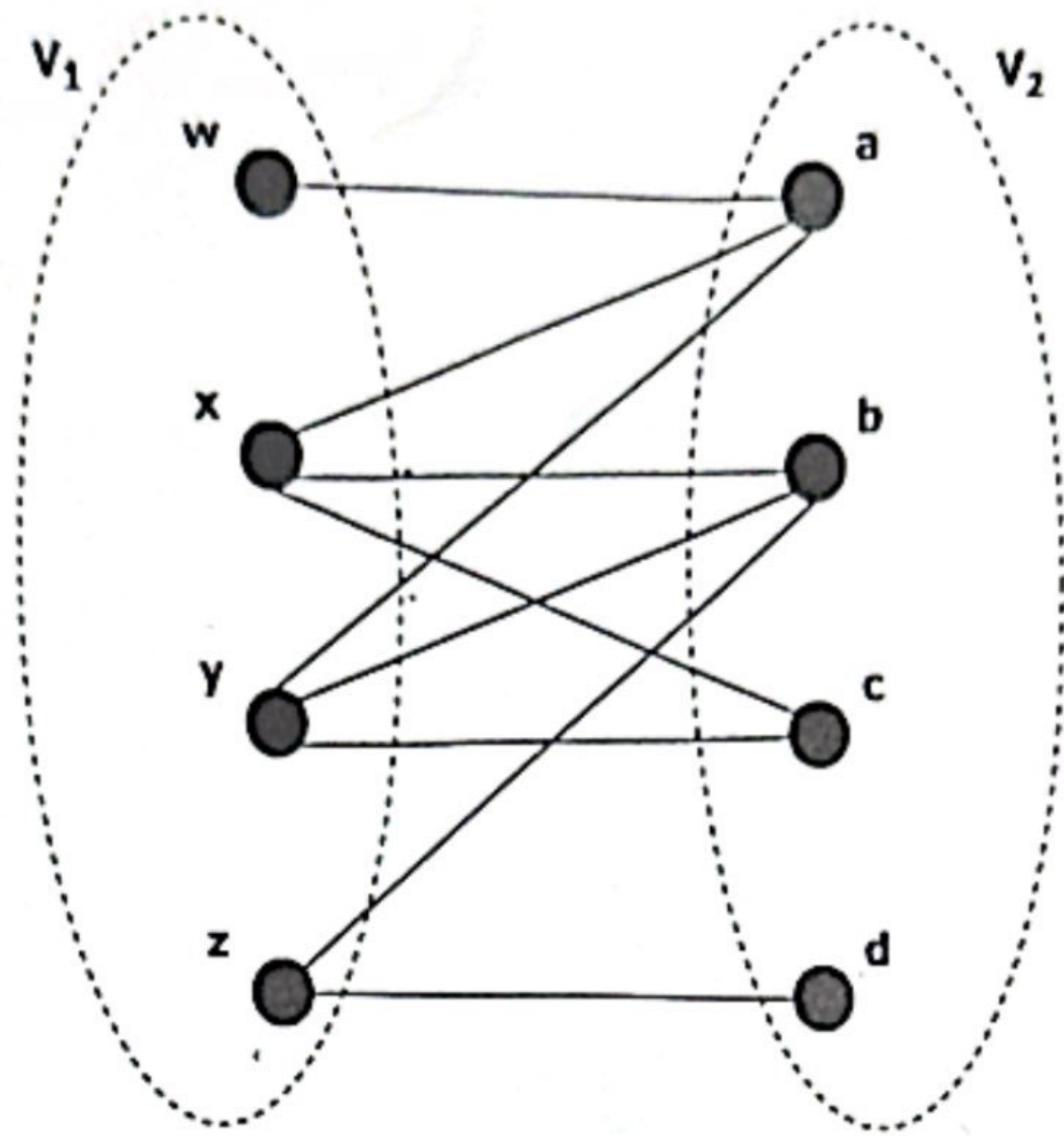


Figure 3: A sample bipartite graph for Question 5.

- a) Given only one eligible applicant can be hired for one position, determine whether it is possible for every applicant to be selected for a position. Show the detailed steps of your process, highlighting the deficiency numbers for each possible nonempty subset of V_1 .

12
(CO2)
(PO2)
- b) Recommend a simplified method that provides a sufficient but not necessary condition for checking whether the given graph has a complete matching. Show how the simplified method can be applied for the given bipartite graph.

2 + 4
(CO3)
(PO3)
- c) Is the given graph planar? Justify your answer.

7
(CO1)
(PO1)
6. Suppose you are given the following adjacency matrix representing a digraph G with 5 vertices, labeled 1 through 5.

$$\begin{bmatrix}
 1 & 1 & 0 & 0 & 0 \\
 0 & 0 & 1 & 0 & 1 \\
 1 & 0 & 0 & 1 & 1 \\
 0 & 0 & 0 & 0 & 0 \\
 1 & 0 & 0 & 1 & 0
 \end{bmatrix}$$

- a) Differentiate between adjacency matrix and incidence matrix representations of graphs. Find the incidence matrix representation of G , assuming appropriate labels for the edges.

5 + 6
(CO1)
(PO1)
- b) Find the number of distinct directed edge sequences of length 3 from vertex 2 to vertex 1 in the digraph G .

6
(CO1)
(PO1)
- c) Find the chromaticity and the geometric dual of the undirected graph corresponding to G .

4 + 4
(CO1)
(PO1)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 150

CSE 4807: IT Organization and Management

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. At BrightTech Ltd., the management team met to plan the company's future. They decided on different corporate strategies: expanding into software (growth), keeping the hardware division steady (stability), and fixing issues in customer service (renewal). To achieve this, they used different types of plans. Managers set goals by looking at the company's mission, checking their resources, setting clear and realistic targets. This helped everyone stay focused and work towards the company's success.
- a) In the scenario, different types of plans were used depending on the situation. How can plans be classified based on their breadth, time frame, specificity, and frequency of use? Explain with examples. 6 (CO1) (PO1)
- b) The managers at BrightTech followed a process to set clear and realistic goals. What are the main steps managers should follow when setting goals? 4 (CO1) (PO1)
- c) BrightTech used different corporate strategies like growth, stability, and renewal. What do these corporate strategies mean, and how were they applied in the scenario? 15 (CO1) (PO1)
2. Sarah is a team leader at NovaTech. She changes her leadership style based on her team's experience. For new employees, she tells them what to do, and for skilled workers, she gives them freedom, following Situational Leadership. She believes that good managers must be leaders, but leaders don't always have to be managers. Using the path-goal theory, she supports her team when they are stressed and gives clear steps when tasks are unclear. Instead of using her job title to exercise power, Sarah relies on her skills and the trust of her team to lead well.
- a) Using Hersey and Blanchard's Situational Leadership model, how does Sarah adjust her leadership style based on the teams' readiness or experience levels. 7 (CO1) (PO1)
- b) Sarah believes that all managers should be leaders, but not all leaders need to be managers. Do you agree with this view? Explain why or why not. 5 (CO2) (PO1)
- c) How does Sarah apply path-goal theory in her leadership approach when her team is stressed or facing unclear tasks? Discuss preferred leadership behaviors based on different path-goal situations. 2 + 6 (CO1) (PO1)
- d) What sources of power can Sarah use to lead her team effectively? 5 (CO1) (PO1)

3. At Nova Tech, manager Mr. Hasan notices declining product quality and delivery delays. He reviews performance against standards and adjusts the production process. Focusing on key metrics like defect rates and employee efficiency, he applies control process to catch problems early and measures overall organizational performance. Finance tracks different financial ratios and leadership evaluates overall organizational performance to drive continuous improvement.
- Using the three-step control process, how can a manager identify issues and take corrective actions? 8
(CO1)
(PO1)
 - Why is "what we measure" more important than "how we measure" in the control process? If you had to choose, which control strategy would you prefer among feedforward, concurrent, and feedback control? Justify your choice. 3 + 4
(CO4)
(PO2)
 - What are the popular financial control measures and how are these financial ratios calculated within an organization? 2 + 4
(CO1)
(PO1)
 - What criteria or indicators are commonly used to measure organizational performance? 4
(CO1)
(PO1)
4. a) TechNova Electronics authorizes a new project to develop a faster, cheaper, and smaller laptop based on advances in computer memory and electronics technology. Based on recent innovations in microchips and memory storage, the company has initiated this project to develop a next-generation ultra-portable laptop. The project will proceed through a well-defined life cycle, from initiation to completion, providing the business with both tangible and intangible benefits.
- What categories of the project initiation factors does the above scenario meet? 5
(CO2)
(PO1)
 - Describe the different types of project life cycle with example. 10
(CO1)
(PO1)
 - What is meant by tangible and intangible benefits in a project? 4
(CO1)
(PO1)
- Suppose that a project is estimated to be 400 KLOC. Calculate the effort and development time for each of the three modes of COCOMO model, i.e., organic, semidetached, and embedded. 6
(CO2)
(PO2)
5. TechNova wanted to work faster and better. They learned that there are two types of structures: mechanistic and organic. They put a great deal of thought into designing an appropriate structure that depends on some contingency variables or factors. They grouped workers by departmentalization following five common ways. To remain flexible, TechNova started using team-based groups to help employees work together and be more creative.
- How do the two main types of organizational designs differ TechNova learned about? 5
(CO1)
(PO1)
 - What are the contingency factors and how did TechNova use them in designing their structure? 6
(CO1)
(PO1)

c) What is departmentalization and what are the five common ways TechNova grouped their employees? Give a simple example for each.

8
(CO2)
(PO1)

d) What kind of new or contemporary organizational design did TechNova adopt to be more flexible and creative?

6
(CO1)
(PO1)

6. A project network data is given in Table 1.

Table 1: Data for Question 6.b and 6.c.

Task	A	B	C	D	E	F	G	H	I	J	K	L	M
Most optimistic time	2	3	1	2	4	2	3	4	3	2	1	2	1
Most likely time	4	5	2	3	6	4	5	6	5	3	2	4	3
Most pessimistic time	6	9	3	8	8	6	9	10	7	6	5	6	5
Predecessors	-	-	A	A	B	B	C, D	E	F	G	H, I	J, K	L

a) What is the difference between PERT and CPM?

4
(CO1)
(PO1)

b) Draw the network diagram based on the given precedence relationships and calculate the expected time and variance for each activity using Table 1.

6
(CO3)
(PO2)

c) Using data from Question 6.b, find the critical path, float time after calculating the earliest start (ES) and latest start (LS) time and earliest finish (EF) and latest finish (LF) time. Draw the final network diagram according to your result.

15
(CO3)
(PO2)